Set 1 Answer

Scenario based question

1. Healthcare Startup

Logic

1. Load the diabetes dataset.
2. Separate the input features and target variable.
3. Clean and preprocess the data.
4. Apply a feature selection technique such as SelectKBest or RFE.
5. Select only the most important features.
6. Train the machine learning model using the selected features.
7. Evaluate the model performance.

2. Student Portal

Logic

1. Create a Django project named student_portal.
2. Create a Django application.
3. Register the app in settings.py.
4. Create database models for student information.
5. Run migrations.
6. Create views and URLs.
7. Design HTML templates.
8. Run the server and test the application.

3. Customer Purchase Pattern

Logic

1. Load the customer dataset.
2. Perform preprocessing.
3. Standardize the features.
4. Apply PCA to reduce dimensionality.
5. Select principal components that retain most of the information.
6. Train the prediction model using the reduced dataset.
7. Evaluate model accuracy.

4. Digital Library Recommendation System

Logic

1. Collect user-book rating data.
2. Create a User–Book rating matrix.
3. Calculate similarity between users.
4. Find users with similar reading interests.
5. Identify books liked by similar users.
6. Remove books already read by the target user.
7. Recommend the remaining books.

5. Credit Risk Assessment

Logic

1. Load the banking dataset.
2. Clean missing values.
3. Separate features and target.
4. Apply RFE or SelectKBest to identify important financial indicators.
5. Remove less important features.
6. Train the classification model.
7. Predict customer risk level.

6. News Recommendation

Logic

1. Collect user reading history.
2. Extract article features (category, keywords).
3. Build a content-based recommendation model.
4. Build a collaborative filtering model.
5. Combine both recommendations.
6. Rank the recommended articles.
7. Display the best recommendations.

7. Spam Detection

Logic

1. Collect email dataset.
2. Convert emails into numerical features using TF-IDF.
3. Apply feature selection (Chi-Square or SelectKBest).
4. Keep only the most relevant text features.
5. Train the spam classifier.
6. Predict whether new emails are spam or not.

8. Course Recommendation

Logic

1. Collect learner course enrollment data.
2. Create a learner course matrix.
3. Calculate learner similarity.
4. Find learners with similar interests.
5. Identify courses enrolled by similar learners.
6. Remove courses already completed.
7. Recommend new courses.

9. Car Price Prediction

Logic

1. Load the car dataset.
2. Clean and preprocess the data.
3. Standardize numerical features.
4. Apply PCA to reduce the number of features.
5. Train the regression model.
6. Predict the car price.

10. Product Recommendation for New Users

Logic

1. Identify that the user has no previous interaction .
2. Recommend popular or trending products.
3. Ask the user to select preferences if possible.
4. Recommend products based on selected categories.
5. As the user interacts, switch to personalized recommendations.